

6 Malaria is a life-threatening disease caused by the pathogen *Plasmodium*.

Plasmodium is transmitted into the blood plasma of a person through the bite of an infected mosquito.

The *Plasmodium* is a sporozoite that travels to the liver cells.

This sporozoite produces merozoites that infect red blood cells.

(a) State the meaning of the term **infection**.

(1)

(b) In 2019, there were an estimated 229 million cases of malaria worldwide.

The number of deaths was estimated at 4.09×10^5 per year.

(i) Which is the approximate death rate from malaria in 2019?

(1)

- ☐ **A** 1 in 180
- ☐ **B** 1 in 600
- ☐ **C** 1 in 1 800
- ☐ **D** 1 in 6 000

(ii) In 2019, 274 000 young children died from malaria.

Which is the ratio of deaths in young children to deaths in other age groups?

(1)

- ☐ **A** 0.7:1.0
- ☐ **B** 1.0:0.5
- ☐ **C** 1.0:1.5
- ☐ **D** 2.0:1.0

(c) Scientists are trying to develop vaccines to protect against malaria.

One vaccine uses living sporozoites.

(i) Suggest why sporozoites are used in a vaccine against malaria.

(3)

- (ii) In clinical trials of this vaccine, people were injected with the vaccine and two chemicals.

One chemical kills sporozoites and the other kills merozoites.

Deduce why these two chemicals were injected with the vaccine in these clinical trials.

(3)

- (iii) Later in the clinical trials, people were infected with one of two strains of sporozoites, three months after the vaccination.

One strain was identical to the one used in the original vaccine and the other strain was a different one.

The trials found that the vaccine was effective against both strains.

Explain these findings.

(3)

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(Total for Question 6 = 12 marks)